Section 1.0: Document Control & Revision 2.0 Delta

Document ID: HLD-PT-2026-002-REV2 | Revision 2.0 (Upgraded Architecture)
Domain: Powertrain Domain Controller (ECU_EMS_01)
Target Release: Electric Vehicle Performance Pack v2.0

Key Architectural Modifications in Revision 2.0:
1. Increased task execution frequency for TorqueArbitrationSWC from 10ms to 5ms for rapid EV torque response.
2. Safety Integrity Level (ASIL) upgraded from ASIL B to ASIL C under ISO 26262.
3. Introduced new Software Component: RegenerativeBrakingSWC (Application SWC, ASIL C).
4. CAN FD E2E Protection updated from Profile P01 to Profile P02 (16-bit CRC, 8-bit alive counter).
5. Added DTC 0x932001 and DID 0x2201 for regenerative deceleration monitoring.

Section 2.0: Powertrain Controller Architecture Chapter 2

Detailed architectural specification content for Powertrain Domain Controller v2.0 (Page 2).

Includes AUTOSAR Classic 4.4.0 configuration specifications, port interfaces, and ISO 26262 ASIL C verification criteria.

Section 3.0: Powertrain Controller Architecture Chapter 3

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Section 4.0: Powertrain Controller Architecture Chapter 4

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Section 5.0: Powertrain Controller Architecture Chapter 5

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Section 6.0: TorqueArbitrationSWC Specification (Revision 2.0)

6.1 Modified TorqueArbitrationSWC Architecture:
- Periodic Rate: 5ms (Accelerated from 10ms in v1.0; mapped to OsTask_5ms_Core1).
- Safety Integrity Level: ASIL C (Upgraded from ASIL B).
- New Required Port: Rp_RegenTorqueLimit (R-Port) -> Interface: If_RegenTorque.
  * Data Element: MaxRegenTorque_Nm (sint16), RegenAvailable (boolean).
- Port Pp_TorqueCommand: Transmits arbitrated motor torque at 5ms update cycle.

Section 7.0: RegenerativeBrakingSWC Specification (New in v2.0)

7.1 Component Purpose (Newly Introduced Component in v2.0):
RegenerativeBrakingSWC calculates maximum permissible regenerative deceleration torque
based on battery state-of-charge (SoC), cell temperature, and anti-lock braking (ABS) slip thresholds.

7.2 Scheduling and Safety:
- Periodic Rate: 5ms (mapped to OsTask_5ms_Core1).
- ASIL Classification: ASIL C.

7.3 Ports and Interfaces:
- Port Rp_BatteryState (Required): Interface If_BatteryStatus.
  * Data Elements: BatterySoC_pct (uint8), MaxChargePower_kW (uint16).
- Port Pp_RegenTorqueLimit (Provided): Interface If_RegenTorque.
  * Data Elements: MaxRegenTorque_Nm (sint16), RegenAvailable (boolean).

Section 8.0: Powertrain Controller Architecture Chapter 8

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Section 9.0: Powertrain Controller Architecture Chapter 9

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Section 10.0: Powertrain Controller Architecture Chapter 10

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Section 11.0: Powertrain Controller Architecture Chapter 11

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Section 12.0: Powertrain Controller Architecture Chapter 12

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Section 13.0: Powertrain Controller Architecture Chapter 13

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Section 14.0: E2E Profile P02 Protection (Upgraded in v2.0)

14.1 E2E Profile P02 Specification:
In Revision 2.0, communication protection is upgraded to E2E Profile P02:
- 16-bit CRC computed using CCITT polynomial (0x1021).
- 8-bit Alive Counter (0..255) for enhanced transmission monitoring on 5ms cycle times.

Section 15.0: Powertrain Controller Architecture Chapter 15

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Section 16.0: Powertrain Controller Architecture Chapter 16

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Section 17.0: Powertrain Controller Architecture Chapter 17

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Section 18.0: Powertrain Controller Architecture Chapter 18

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Section 19.0: Powertrain Controller Architecture Chapter 19

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Section 20.0: Powertrain Controller Architecture Chapter 20

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Section 21.0: Powertrain Controller Architecture Chapter 21

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Section 22.0: Powertrain Controller Architecture Chapter 22

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